

Network Analysis and Engineering

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Introduction

Mathematics is a study of the properties, relationships, and structures of quantities, numbers, shapes, and patterns. Frequently referred to as the universe's language because it provides a systematic means of describing and comprehending the world around us. It requires logical reasoning and critical thinking to solve problems and make hypotheses about the natural world. Several fundamental branches of mathematics comprise algebra, geometry, calculus, probability theory, and statistics.

To dive deeper into Mathematics, we are going to be focusing on Linear Algebra. Linear Algebra is a branch of mathematics that deals with vectors, matrices, finite or infinite dimensions, and linear mappings between such spaces. It is used in pure and applied mathematics, as well as other technical fields such as physics, engineering, and natural sciences.

Linear Algebra in Engineering provides a solid and versatile mathematical framework for the analysis, modeling, and resolution of challenging issues that arise in a range of engineering fields. It is essential to disciplines including robotics, control systems, electrical circuits, signal processing, structural analysis, computer graphics, optimization, and machine learning. In particular, we have Network analysis which uses mathematical tools in studying interconnected systems that assists engineers in diagnosing problems, improving performance, designing effective solutions, and understanding the relationships between systems.

Network Analysis is a branch of mathematics that involves the study of interconnected systems represented as graphs. Matrices and vectors are usually used to represent the relationships between components in a network. Its key concepts include analyzing connectivity patterns, identifying important nodes through eigenvector centrality, and solving systems of linear equations to study network flows. In addition, network analysis is widely used in many different areas, such as transportation systems, communication networks, and electrical circuits, and it offers a mathematical framework for the analysis and optimization of intricate networks.

Furthermore, we have Kirchhoff's Laws assert that the total accumulation of voltage around a loop equals the aggregate reduction in voltage. The Electrical Network, is the interconnected arrangement of electrical components or a model representing such interconnections, including diverse electrical elements. This concept involves the examination of how these elements are connected to one another. There's also Traffic Flow, where we use math to describe how cars, drivers, and roads interact. All of this helps us see how Linear Algebra connects to Network Analysis and Engineering.

4.3 A section intended for the topics of linear algebra used in the research

Kirchhoff's Laws

In 1845, the electrical engineering landscape underwent a transformative change with the pioneering work of German physicist Gustav Kirchhoff. His seminal contributions created two fundamental laws that have become essential pillars in the field of electrical engineering: Kirchhoff's Current Law and Kirchhoff's Voltage Law. These laws, derived from Kirchhoff's extensive knowledge, have evolved to play a vital role in the analysis and understanding of electrical circuits, forming the basis of our understanding of current and voltage interactions in diverse applications across modern technology and infrastructure. Kirchhoff's Current and Voltage Laws are two important laws of circuit analysis and serve as the foundation for network analysis.

Current law states that the total current flowing into a junction or node precisely equals the current flowing out of the node. In other words, the algebraic sum of currents converging at a node in a conductive network is zero. This law of Kirchhoff is known as the conservation of charge. Consider currents i_1 and i_2 as incoming flows to a node, while i_3 and i_4 represent outgoing currents. Consequently, i_1 and i_2 constitute one side of the equation, while i_3 and i_4 form the other side. This law highlights the fundamental concept that, within an electrical circuit, the total charge remains conserved at any given node, providing an essential structure for understanding and analyzing current distribution in complex networks.

Voltage law states that the total voltage around a closed loop in an electrical circuit equals the sum of all voltage drops within that same loop. This law of Kirchhoff is known as the conservation of energy. When determining the voltage polarities of passive elements, a convention is employed: the terminal where the current enters the circuit element is deemed positive, while the terminal where it exits is considered negative. Maintaining a consistent direction in the chosen loop is critical for accurate analysis. Whether the cycle is established as clockwise or anticlockwise, this direction must be steadfastly maintained to ensure that the final sum of voltages adds up to zero. Any deviation from this chosen direction could compromise the accuracy of the analysis, as the final sum of voltages may no longer equate to zero. Therefore, adherence to this directional consistency is imperative for the effective application of Kirchhoff's Voltage Law, ensuring precise circuit analysis and a comprehensive understanding of energy distribution within the system.

Electrical Network

Electrical networks are an interconnection of different electrical components (active and passive) in a specified manner to form a system. These electrical components each have a specific function. The parts of an electrical network include a load or any device that consumes power. This may be a form of an appliance, motors, and such. To power the network, it needs an electrical source, whether it be a generator or a battery. It also includes controlling devices such as switches and circuit breakers to control the electricity's flow intensity. A network would also require protection devices to mitigate the damages done by a short circuit or other electrical

issues. Electric fuses and switchgear systems are an example of these. It also needs a conductive medium so that the current can flow from an energy source and back.

An electrical network can be categorized into different types based on their applications. If a network has a closed path, then it is called a circuit. Circuits are the most common type of network, since it builds foundation on several modern technologies such as smartphones, television, and speakers. Other networks include power distribution networks, communication networks, control systems, and instrumentation networks.

Traffic Flow

In engineering, traffic flow is the study of the movement of drivers, vehicles and the interactions that they make with each other with the goal of understanding and developing a network that results in an efficient and smooth traffic flow. The mathematical theory behind traffic flow first began in the 1920s when Frank Knight produced an analysis of traffic equilibrium. However, despite the development of technology, there have been no general theories that could be applied to the real flow conditions. The traffic models developed use a mixture of theoretical and empirical techniques, which are then used to create traffic forecasts. These give information about minor and major changes such as, increased vehicle use, changes in land use and mode of transport. Additionally, this provides data which can identify areas of congestion, where the entire road network has to be adjusted.

Mathematical models of traffic flow generally describe a system by a set of variables and equations. Mathematical modeling of traffic flow follows the Junction Rule, where at each junction in the network, the total number of vehicles flowing in must be equal to the total number of vehicles flowing out. The models created are a simplified representation of the real-world system, which help give an understanding and formulate possible solutions for the traffic problem.

System of Linear Equations

A system of linear equations is a set of m linear equations that has n unknowns. This can be represented as a matrix equation in the form $Ax = b$, or in the an augmented matrix in the form $[A|B]$. The matrix A contains the coefficients of the variables, while B contains the constants of the linear system. The column matrix x represent the variables of the system. A linear system may either be consistent or inconsistent. A system is consistent if it has a unique solution or an infinite number of solutions. Meanwhile, a system is inconsistent if it has no solutions.

In the context of traffic flow, a system of linear equations is used to represent the number of vehicles flowing in and out of the traffic network. The values may be unknown or may be represented by a variable $x_i, i \in Z$. The values and variables represent the number of vehicles per unit of time passing through a network of roads.

Gauss - Jordan Elimination

The Gauss - Jordan elimination, named after Carl Friedrich Gauss and Wilhelm Jordan, is a method that continues the reduction process of the Gaussian elimination to obtain the reduced row echelon form of a matrix. This method is an algorithm that uses a sequence of row operations to solve a system of linear equations. Row operations are performed in a matrix to obtain an easier to solve equivalent system of linear equations.

The system that was obtained from the data is solved through the Gauss - Jordan elimination. This predicts the number of vehicles that should pass through each junction to ensure a smooth traffic flow.

4.4 A section about detailed discussion of the use of linear algebra in the assigned field

Kirchhoff's Laws

Linear algebra serves as a fundamental mathematical framework in the analysis of electrical circuits, particularly in the application of Kirchhoff's laws. These laws, Kirchhoff's Current Law (KCL) and Kirchhoff's Voltage Law (KVL), are foundational principles governing the behavior of current and voltage distributions in complex circuits. The integration of linear algebraic methods enhances the efficiency and systematic approach to solving these intricate electrical systems.

In the context of Kirchhoff's Current Law, which states that the algebraic sum of currents into any junction is zero, linear algebra plays a crucial role in formulating and solving systems of equations derived from nodal analysis. Nodal analysis, a prevalent method in circuit analysis, expresses currents at different nodes as a system of linear equations. By representing these equations in matrix form, linear algebra facilitates a systematic and efficient analysis, particularly in circuits with multiple nodes. The application of matrices and vectors provides a structured representation of the nodal currents, simplifying the mathematical model of the circuit.

Similarly, Kirchhoff's Voltage Law, describing the distribution of voltage within a closed loop of an electrical circuit, relies on linear algebra for solving equations derived from loop analysis. Loop analysis expresses voltage drops across various elements in a closed loop as a system of linear equations. Matrices come into play by providing a concise representation of these equations, making it easier to analyze circuits with multiple loops. The systematic approach afforded by linear algebra ensures clarity and efficiency in dealing with complex circuit configurations.

Electrical Network

From simple networks to complex ones, the current can be determined by applying Kirchhoff's Current Law and Gauss-Jordan Elimination. To do this, a network equation must be constructed. The total current entering a circuit junction should be the same as the total current leaving, according to KCL. Thus, the net current that entered and left the circuit must be 0. Once a system of equations is formed, a matrix can be constructed from this. Using Gauss-Jordan

Elimination would provide the matrix in reduced row echelon form. This will give the individual currents of the circuit.

While using physics formulas can provide the same data, using matrices would be more efficient. Inverse matrices, Gauss-Jordan elimination, and other linear algebra equations can be used to determine currents. It can also be used to find node values, integrating power switches, accommodating multiple batteries and capacitors, and structuring equations to solve for unknown variables. The linear nature of the current flow model is directly attributed to the linearity of Ohm's Law and Kirchhoff's Law. According to Ohm, the voltage drop across a resistor is directly proportional to the current flowing through it, while Kirchhoff asserts that the sum of voltage drops in a loop equals the sum of voltage sources in that loop. In real-world scenarios, electrical networks often comprise numerous resistances and circuits. To calculate currents across branches, solving extensive systems of equations becomes necessary, a task best suited for computational methods and computers.

Traffic Flow

The analysis of traffic flow within a network of streets involves a detailed mathematical model using a system of linear equations, and solving them through the Gauss - Jordan elimination. This method is used to accurately calculate vehicular movement throughout the entire road network. In the model, the system of linear equations serves as a representation of the number of vehicles flowing in and flowing out of the whole network at each intersection. The measured values, which are usually calculated in vehicles per hour (*vph*) may be known or be denoted as variables $x_i, i \in Z$. The derived system of linear equations is then converted into its augmented matrix form. The values of this matrix correspond to the coefficients of each variable and the constants of the equation.

Subsequently, the Gauss - Jordan elimination is used, transforming the augmented matrix of the system into its reduced row echelon form. This technique makes it easier to solve complicated systems of equations by getting rid of variables and simplifying the equations step by step. The solutions obtained from this process may include one or more parameters. These are interpreted as positive integers, as traffic flow is generally represented as a one-way path. The resulting values serve as estimations for the expected number of vehicles at every point of the network. These values serve as information to formulate strategies to optimize traffic flow, minimize congestion, and improve transportation efficiency.

4.5 Summary and conclusion

In summary, linear algebra plays a crucial role in understanding and applying Kirchhoff's Laws in electrical circuits, as well as in the analysis of Electrical Networks and Traffic Flow.

Kirchhoff's Current Law (KCL) and Kirchhoff's Voltage Law (KVL) are fundamental principles in electrical engineering, providing a basis for analyzing current and voltage interactions. Linear algebra, involving matrices and systems of linear equations, enhances the efficiency of solving complex electrical circuits governed by these laws. This is particularly evident in nodal and loop analysis, where linear algebra provides a structured representation, simplifying the mathematical models of circuits.

In Electrical Networks, linear algebra operations, such as Kirchhoff's Current Law and Gauss-Jordan Elimination, are employed to determine currents within the network. The application of matrices proves more efficient than traditional physics formulas, especially in scenarios with extensive networks and multiple components. Linear algebra aids in solving systems of equations, finding node values, and accommodating various components within electrical networks.

The importance of linear algebra extends to Traffic Flow analysis, where a system of linear equations models vehicular movement in a network of streets. The Gauss-Jordan elimination method is applied to accurately calculate traffic flow throughout the road network. Linear algebraic methods, including the representation of equations in matrix form, facilitate a systematic and efficient analysis, contributing to strategies for optimizing traffic flow and improving transportation efficiency.

Conclusively, the integration of linear algebraic methods provides a powerful and systematic approach across Kirchhoff's Laws, Electrical Networks, and Traffic Flow, demonstrating the interdisciplinary nature and broad applicability of these mathematical tools in diverse engineering applications.

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